Amabwiriza y'umuyagankuba: sitidiyo

Ibyo umuyagankuba (electrician) akeneye byose ngo ashyire amashanyarazi muri sitidiyo imwe ya kontineri: first fix mbere y'uko inkuta zifungwa, second fix, igerageza (tests) n'icyemezo.

> ***Ubusobanuro:*** *iyi nyandiko yahinduwe na mudasobwa (machine draft). Igomba gusuzumwa n'umuntu uvuga Ikinyarwanda kavukire mbere yo kuyiha abakozi. Iyo hari itandukaniro, kurikiza iy'Icyongereza.*

Status: verisiyo A2, 2026-10-28, PRELIMINARY. Ibipimo byose by'amashanyarazi (ratings) ni ibyo igishushanyo giteganya gusa (design intent), kugeza igihe umuyagankuba (electrician) ufite uruhushya rwa RURA (class A) azategura installation, akayishyiraho, kandi akayiha icyemezo (certificate) hakurikijwe RS 565-1:2011.

Sitidiyo zombi zishyirwamo amashanyarazi kimwe, bityo aya mabwiriza akora kuri zombi.

English: read this guide in English · Gucapa: aya mabwiriza nka PDF

Amagambo akoreshwa hano

- **End A (impera A)**: urukuta rufunze rw'impera, iruhande rw'igitanda. **End B (impera B)**: impera ifite inzugi za kontineri, iruhande rw'ubwiherero.
- **Back wall (urukuta rw'inyuma)**: urukuta rurerure rudafite aho gufungukira. **Front wall (urukuta rw'imbere)**: urukuta rurerure rufite urugi runyerera rw'ikirahure.
- Imyanya iri mu mbonerahamwe ya points ni milimetero zibarwa uhereye ku rukuta rurangiye rwa end A (x), uhereye ku rukuta rw'inyuma rurangiye (y), na hejuru uhereye hasi harangiye (z).

Sitidiyo mu ncamake

Iki	Igishushanyo (design)
Aho umuriro uturuka (supply)	inlet imwe ya 32 A (aho umuriro winjirira), IP67, IEC 60309 y'ubururu (2P+E), hanze ku rukuta rw'inyuma, inyuma neza ya board; isoko y'umuriro iyo ari yo yose ya 230 V
Breaker nkuru (main)	breaker (disjoncteur) double-pole ya 32 A (C curve) na DIN kWh check meter (mubazi yo kugenzura); tails za 6 mm² zinyura muri gland

Iki	Igishushanyo (design)
Board (tableau)	imyanya 24 (modules) mu mirongo ibiri ya 12: breaker nkuru, check meter, SPD type 2, priority relay na contactor yayo, RCBO imwe (30 mA, type A) kuri buri muzunguruko (circuit)
Imizunguruko (circuits)	8, kuva C1 kugeza C8 (imbonerahamwe iri hepfo)
Umuriro ntarengwa ukenewe (maximum demand)	30.0 A (diversified): amatara 0.3 + fans 0.3 + priza (sockets) 16.4 + ikinini muri ibi bibiri: hob 11.6, cyangwa heater 8.7 hamwe na AC 4.3
Earthing (guhuza n'ubutaka)	TT: earth rod (piquet de terre) imwe kuri buri kontineri; ntuzigere ukoresha umuriro (supply) uhuza neutral na earth (PME)
Inzira y'insinga (cable route)	conduit ya mm 20 mu myanya iri hagati y'ibiti (battens), itwikirwa na spray foam nyuma ya first fix

Ryari	Iki	Intambwe
Nyuma y'ibiti (battens, S03)	First fix: earth rod, conduits, udusanduku (boxes), insinga	S04
Nyuma ya lining n'igikoni	Second fix: board, ibikoresho (devices), igerageza, icyemezo (certificate)	S19

Ingingo eshatu zo guhagarara ukagenzura (hold points)

1. **Ibiti bibanza.** Tangira first fix gusa iyo ibiti (S03) bimaze gufatwa kandi impapuro za hasi zimaze gushyirwaho.
2. **Mbere ya foam.** Buri conduit, buri gasanduku na buri nsinga bigomba kuba byashyizwemo, byakorewe igerageza ry'insulation (insulation test), byafotowe urukuta ku rundi, kandi byasinyweho mbere ya spray foam (S05). Foam itwikira conduits: nyuma nta kindi gishobora kongerwamo, keretse ukurura insinga uzinyujije muri conduits.
3. **Mbere yo guha umuriro.** Igerageza ritsinda, kandi icyemezo cya RURA (certificate) gisinywa, mbere y'uko inlet ihabwa umuriro bwa mbere.

S04: first fix (imirimo ya mbere)

1. Shyira agasanduku ka board ku rukuta rw'inyuma aho igikoni kiri, n'inlet hanze, inyuma yako neza: ni imyanya DB na PI mu mbonerahamwe ya points.
2. Shinga earth rod (E1) hanze ya end B, ukurure insinga yayo uyigeze kuri earth bar ya board; resistance yayo uzayipima nyuma, mu gihe cy'igerageza.

3. Zamura conduit ya mm 20 uhereye kuri board ujya mu mwanya uri mu gisenge (ceiling void), uyinyuze muri uwo mwanya, hanyuma uyimanure ugororotse ujya kuri buri gasanduku. Nta conduit ihengamye (diagonal); fata buri gasanduku ku giti (batten).
4. Kurura insinga za buri muzunguruko (circuit): 1.5 mm² kuri C1 na C6, 2.5 mm² kuri C2, C3, C5, C7 na C8, 4 mm² kuri C4.
5. Sohora C6 (amatara yo hanze) unyujije muri gland imwe iri ku gice cy'urukuta rw'imbere kidafite ikirahure (solid front strip), ku ruhande rw'igitanda; hanze, ukomeze muri conduit itangizwa n'izuba (UV-stable).
6. Ubwiherero: itara rya IP65 na fan isohora umwuka (extract fan) ya IPX4 (zone 1, hejuru ya tray kandi munsi ya m 2.25), nta priza, kandi switch yabyo iri hanze y'urugi.
7. Andika nimero (id) ya buri point ku cyuma iruhande rw'agasanduku kayo, kugira ngo umubaji akate impapuro za lining ahantu nyaho.
8. Gerageza continuity n'insulation resistance kuri buri nsinga, ufotore buri rukuta, hanyuma ni bwo usinya uruhushya rwo gushyiramo foam (release).

Imizunguruko (circuits): insinga, breaker, uburebure bw'insinga n'igabanuka rya voltage (voltage drop) · download CSV

circuit	serves	device	cable_mm2	design_A	derated_Iz_A	run_m
C1	Indoor lights and fans	B6	1.5	0.6	13.6	34.9
C2	Sockets: bed, sofa and deck	B16	2.5	8	18.3	24.3
C3	Kitchen sockets, fridge and hood	B16	2.5	14	18.3	12.3
C4	Induction hob	B20	4	15.2	24.4	4.4
C5	30 L water heater	B16	2.5	8.7	18.3	3.9

circuit	serves	device	cable_mm2	design_A	derated_Iz_A	run_m	
C6	Outdoor lights: deck, stair and roof	B6	1.5	0.2	13.6	16.5	
C7	AC provision	B16	2.5	12	18.3	3.9	
C8	Spare	B16	2.5		18.3		

Points: icyo ari cyo, aho kiri n'uburebure bwacyo kuva hasi · download CSV

point	kind	circuit	description	x_mm	y_mm	z_mm	watts	face
DB	db		Unit board, 24 modules (2 x 12), by the galley	3678	0	1625	0	+y
PI	inlet		Power inlet 32 A IP67 (IEC 60309 blue, 2P+E), outside behind the board	3678	-60	1225	0	-y
L1	light	C1	Ceiling light, bed	800	1126	2316	12	-z
L2	light	C1	Ceiling light, living	2300	1126	2316	12	-z
L3	light	C1	Ceiling light, galley	3778	1126	2316	12	-z
L4	light	C1	Bathroom light IP65	5223	850	2316	12	-z
L5	light	C1	Wall reading light over the bed (wall side)	300	0	1150	4	+y

point	kind	circuit	description	x_mm	y_mm	z_mm	watts	face
L7	light	C1	Under-cabinet LED strip	4278	800	1790	10	-z
S1	switch	C1	Entry: bed (2-way), living and galley lights	4480	2252	1100	0	-y
S2	switch	C1	Bed: ceiling light (2-way), over the bedside table	1810	0	1000	0	+y
S3	switch	C1	Bathroom light + fan (outside)	5368	1770	1100	0	+y
F1	fan	C1	Bathroom extract 100 mm, IPX4 for zone 1, with run-on	4898	0	2100	20	+y
F2	fan	C1	Wall fan 300 mm	1800	0	2050	50	+y
P1	usb	C2	Socket + USB-C in the headboard (wall side)	300	18	750	200	+y
P2	usb	C2	Bedside socket + USB-C (reading lamp)	1810	0	650	200	+y
P3	usb	C2	Socket + USB-C over the sofa arm, for the sofa and the table	3150	0	850	500	+y

point	kind	circuit	description	x_mm	y_mm	z_mm	watts	face
P5	socket	C2	Deck socket IP66, outside by the slider	4680	2352	400	300	+y
P6	socket	C3	Worktop double socket 1	4578	275	1100	2000	-x
P7	socket	C3	Worktop double socket 2 (back wall)	4278	0	1100	1000	+y
P8	socket	C3	Fridge socket	4578	275	250	100	-x
P12	socket	C3	Double socket above the hood: hood + Wi-Fi router	4578	850	1720	120	-x
P9	outlet	C4	Hob outlet plate, isolated at its RCBO on the board	4578	850	600	3500	-x
P10	outlet	C5	Heater DP switch + timer	4578	1540	1700	2000	-x
L8	light	C6	Deck light IP65, under the veranda canopy	2900	2395	1984	10	-z
S5	switch	C6	Deck light, beside the entry switch	4575	2252	1100	0	-y
S4	switch	C6	Stair and roof lights (2-way), IP65 at the stair foot	-200	2925	1100	0	+x

point	kind	circuit	description	x_mm	y_mm	z_mm	watts	face
L10	light	C6	Stair light IP65	-900	2925	1500	8	+y
S6	switch	C6	Stair and roof lights (2-way), IP65 at the stair head	-105	2195	3579	0	-x
L9	light	C6	Roof terrace lights IP65 (x2), under the front top rail	1500	2285	3579	20	-z
P11	outlet	C7	AC provision: isolator	900	0	2150	1000	+y
AL1	alarm		Smoke alarm, 10-year battery	1550	1126	2316	0	-z
AL2	alarm		Heat alarm, kitchen, 10-year battery	3428	1689	2316	0	-z
E1	earth		Earth rod 1.5 m copper-bonded in an inspection pit, behind the board	3678	-450	-700	0	+z

S19: second fix (imirimo ya kabiri)

1. Shyiraho ibigize board: breaker nkuru (main breaker) na check meter, SPD, priority relay na contactor, na RCBO zose (B6 kuri C1 na C6, B16 kuri C2, C3, C5, C7 na C8, B20 kuri C4).
2. Shyiraho ibikoresho (devices): amatara, fans, priza, aho hob icomekwa (hob outlet P9: umuriro wayo ucibwa kuri RCBO yayo muri board, igaragara uri kuri hob), n'alarms z'umwotsi n'ubushyuhe (AL1, AL2).
3. Huza (bond) na main earthing terminal: kontineri ubwayo, deck y'imbere, esikariye n'ibyuma byo ku gisenge (roof steelwork).

4. Genzura priority relay: iyo hob ikora, water heater (C5) na AC (C7) bigomba kuzima.

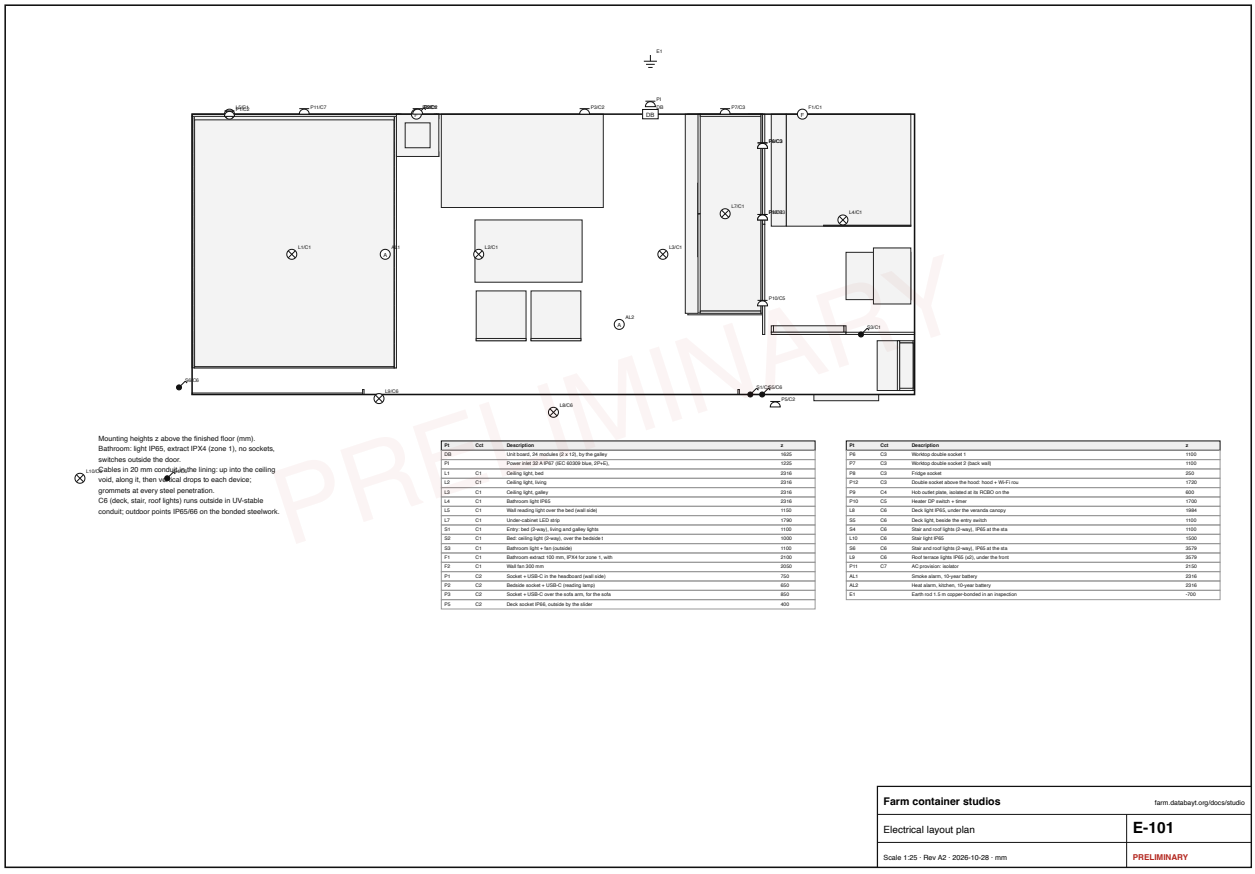
Igerageza mbere yo guha umuriro

1. Continuity ya buri nsinga irinda (protective conductor).
2. Insulation resistance ya buri muzunguruko.
3. Polarity kuri buri priza no kuri buri tara.
4. Resistance ya earth electrode E1: andika agaciro kabonetse.
5. Igihe buri RCBO imara ngo ihagarike umuriro (trip time).
6. Akadirishya kerekana uko SPD imeze (status window), na priority relay indi nshuro.
7. Icyemezo cya RURA (certificate), mbere y'uko inlet ihabwa umuriro.

Guha inlet umuriro

- Isoko y'umuriro iyo ari yo yose ya 230 V, 50 Hz ishobora guha inlet umuriro (board yo ku kibanza, generator cyangwa inverter y'imirasire y'izuba), niba irinda insinga yayo kuri 32 A cyangwa munsi yaho. Isoko y'umuriro ifamu izakoresha ni ikibazo gisubizwa mu `mabwiriza y'umuriro (power guide)`.
- Mubazi ya kWh muri buri sitidiyo ni mubazi yo kugenzura (check meter) gusa, igamije gukurikirana: amashanyarazi abarirwa mu giciro cy'icyumba.

Ibishushanyo



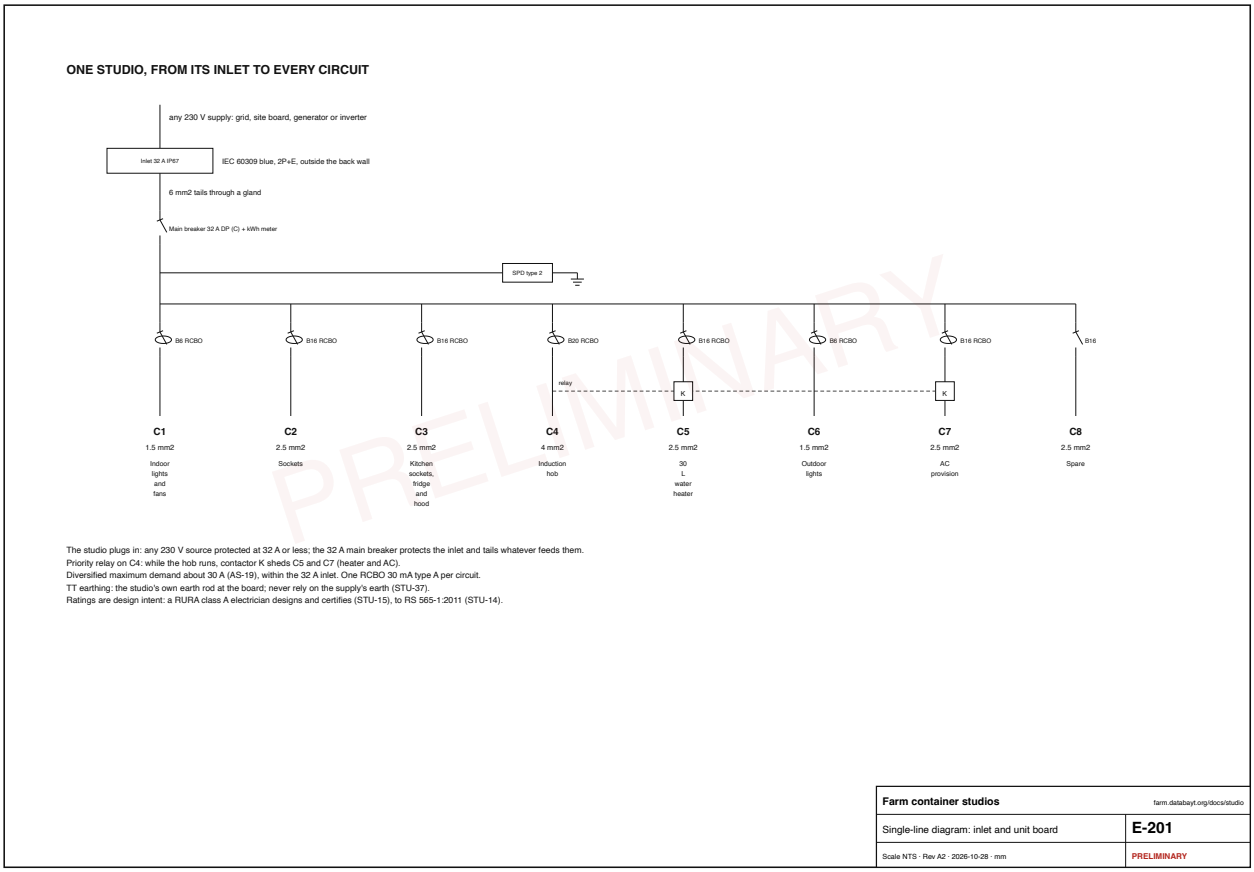
Mounting heights z above the finished floor (mm).
Bathroom: light IP65, extract IPX4 (zone 1), no sockets, switches outside the door.
Cables in 20 mm conduit in the lining: up into the ceiling void, along it, then vertical drops to each device; grommets at every steel penetration.
C6 (deck, stair, roof lights) runs outside in UV-stable conduit; outdoor points IP65/66 on the bonded steelwork.

Pt	Cct	Description	z
DB		Unit board, 24 modules (2 x 12), by the galley	1625
P1		Power inlet 32 A IP67 (IEC 60309 blue, 3P+E),	1235
L1	C1	Ceiling light, bed	2316
L2	C1	Ceiling light, living	2316
L3	C1	Ceiling light, galley	2316
L4	C1	Bathroom light IP65	2316
L5	C1	Wall reading light over the bed (wall side)	1150
L7	C1	Under-cabinet LED strip	1790
S1	C1	Entry / bed (2-way), living and galley lights	1100
S2	C1	Bed: ceiling light (2-way), over the bedside t	1030
S3	C1	Bathroom light + fan (outside)	1100
F1	C1	Bathroom extract 100 mm, IPX4 for zone 1, with	2100
F2	C1	Wall fan 300 mm	2050
P1	C2	Socket + USB-C in the headboard (wall side)	750
P2	C2	Bedside socket + USB-C (reading lamp)	650
P3	C2	Socket + USB-C over the sofa arm, for the sofa	850
P5	C2	Deck socket IP66, outside by the slider	400

Pt	Cct	Description	z
P6	C3	Worktop double socket 1	1100
P7	C3	Worktop double socket 2 (back wall)	1100
P8	C3	Fridge socket	250
P12	C3	Double socket above the hood: hood + Wi-Fi rou	1720
P9	C4	Hob outlet plate, isolated at its RCBO on the	600
P10	C5	Heater DP switch + timer	1700
L8	C6	Deck light IP65, under the veranda canopy	1984
S5	C6	Deck light, beside the entry switch	1100
S4	C6	Stair and roof lights (2-way), IP65 at the sta	1100
L10	C6	Stair light IP65	1500
S6	C6	Stair and roof lights (2-way), IP65 at the sta	3579
L9	C6	Roof terrace lights IP65 (x2), under the front	3579
P11	C7	AC-provision: isolator	2150
AL1		Smoke alarm, 10-year battery	2316
AL2		Heat alarm, kitchen, 10-year battery	2316
E1		Earth rod 1.5 m copper-bonded in an inspection	-700

Farm container studios	farm.databayt.org/docs/studio
Electrical layout plan	E-101
Scale 1:25 · Rev A2 · 2026-10-28 · mm	PRELIMINARY

E-101 Electrical layout plan · open full size



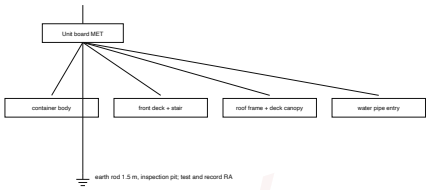
E-201 Single-line diagram: inlet and unit board · open full size

CIRCUIT SCHEDULE (Ib design current, Iz derated capacity, drop at the far point)

Cct	Serves	Device	mm2	Ib A	Iz A	Run m	Drop %
C1	Indoor lights and fans	B6	1.5	0.6	13.6	34.9	0.25
C2	Sockets: bed, sofa and deck	B16	2.5	8.0	18.3	24.3	1.52
C3	Kitchen sockets, fridge and hood	B16	2.5	14.0	18.3	12.3	1.35
C4	Induction hob	B20	4	15.2	24.4	4.4	0.32
C5	30 L water heater	B16	2.5	8.7	18.3	3.9	0.27
C6	Outdoor lights: deck, stair and roof	B6	1.5	0.2	13.6	16.5	0.03
C7	AC provision	B16	2.5	12.0	18.3	3.9	0.37
C8	Spare	B16	2.5	-	-	-	-

Maximum demand item	A
lighting	0.3
fans	0.3
sockets	16.4
hob	11.6
heater	8.7
ac	4.3
total_A	30.0

EARTHING AND BONDING (TT)



earth rod 1.5 m, inspection pit; test and record RA

Bonding: finely stranded green-yellow (16 mm2 [Al]) from each steel item to the MET. RA x 30 mA must stay under 50 V; aim much lower for lightning (measure on site). Tests before handover: continuity, insulation resistance, polarity, earth electrode resistance, RCD trip time, SPD status; certificate from the licensed electrician. Bathroom: zone 1 above the tray to 2.25 m (IPX4); 30 mA on every circuit (STU-38).

PRELIMINARY

Farm container studios	farm.databayt.org/docs/studio
Circuit schedule, earthing and bonding	E-202
Scale NTS · Rev A2 · 2026-10-28 · mm	PRELIMINARY

E-202 Circuit schedule, earthing and bonding · open full size